Probabilistic Ancestral Inference

Reconstructing the Unknown

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CS Capstone

The Missing Link: Reconstructing Stochastic Data

Stochastic — involving random probability and unpredictability where future states cannot be precisely determined.

Why DNA?

DNA follows strict Mendelian inheritance rules (0, 1, 2 copies) and is a real world problem. Missing data comes from lack of financial resources and ethical concerns.

Can we predict the genetics of individuals that are missing in the family tree?

Project Objective:

Simulate, Infer, and Benchmark Genetic Data and Machine Learning Models

The Architecture and Tech Stack

Backend

Python 3.12 and FastAPI

PyTest for Unit Testing

msprime

Frontend

React and Vitest

MUI and RxJS

ML

PyTorch, PyTorch Geometric, hmmlearn, scikit-learn

PyCM, matplotlib, seaborn

Deployment

Docker

Digital Ocean Droplet

The Probabilistic Modeling

Multinomial Logistic Regression

Bayesian Categorical Inference

Hidden Markov Model (HMM)

Deep Neural Network (DNN)

Graph Neural Network (GNN)

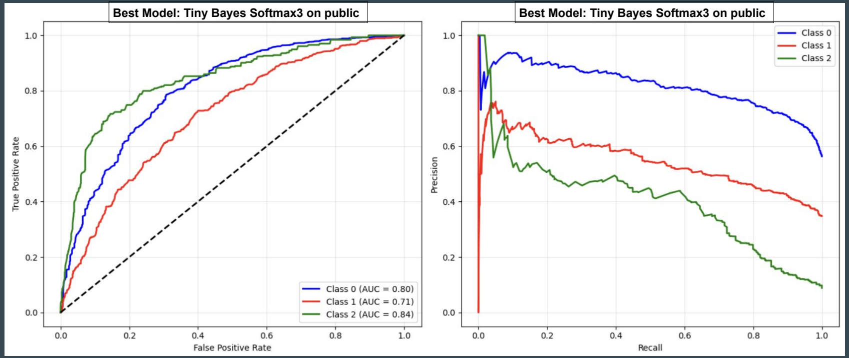
Results and Accomplishments

Bayesian Model Outperforms NN

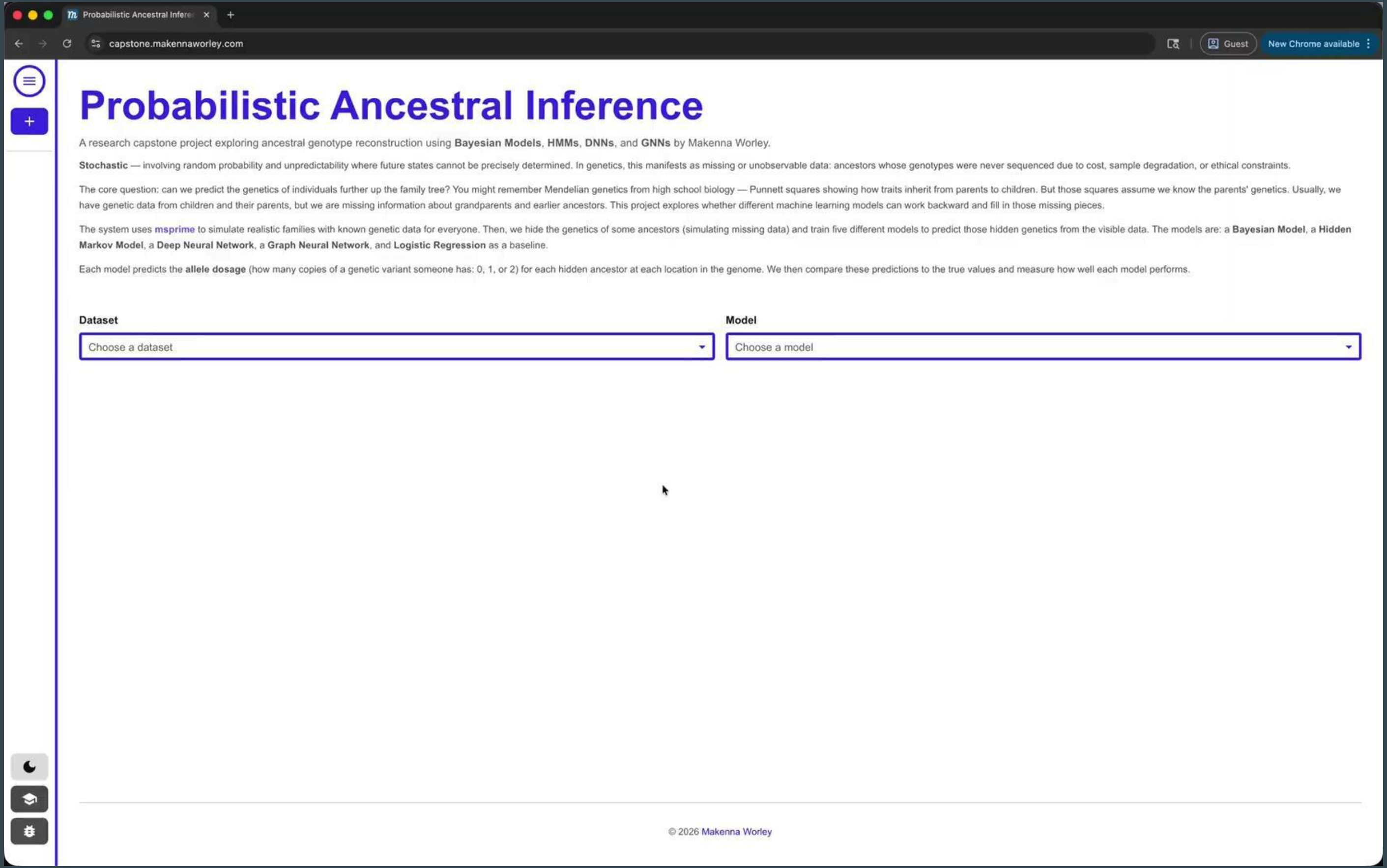
The best model is the Bayesian Inference, it outperform the NN which are statistically worse.

Showing that structured probabilistic reason is more powerful than raw complexity in this case of stochastic data.

Model	Accuracy	Bal. Accuracy	Macro-AUC	F1 Macro	F1 Weighted
tiny, bayes	0.6292	0.6291	0.7857	0.5429	0.6201
tiny, log_reg	0.6243	0.6161	0.7649	0.5337	0.6104
tiny, hmm	0.5527	0.5637	0.7559	0.4406	0.5375
small, dnn	0.521	0.5729	0.7197	0.4837	0.5452
tiny, dnn	0.5178	0.5481	0.7255	0.4667	0.5419
small, log_reg	0.4802	0.5567	0.7368	0.4597	0.5041
small, gnn	0.4224	0.474	0.6603	0.3942	0.453
tiny, gnn	0.4063	0.3523	0.5587	0.3399	0.4449
medium, log_reg	0.3861	0.5046	0.7077	0.382	0.3891
medium, dnn	0.3667	0.4966	0.6939	0.3488	0.3266
medium, gnn	0.3151	0.4746	0.6563	0.2875	0.2612
small, hmm	0.3929	0.4952	0.6858	0.3723	0.4321
medium, hmm	0.1692	0.3791	0.5982	0.1651	0.1377



Demo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Guest
New Chrome available
Probabilistic Ancestral Inference
A research capstone project exploring ancestral genotype reconstruction using Bayesian Models, HMMs, DNNs, and GNNs by Makenna Worley.
Stochastic — involving random probability and unpredictability where future states cannot be precisely determined. In genetics, this manifests as missing or unobservable data: ancestors whose genotypes were never sequenced due to cost, sample degradation, or ethical constraints.
The core question: can we predict the genetics of individuals further up the family tree? You might remember Mendelian genetics from high school biology — Punnett squares showing how traits inherit from parents to children. But those squares assume we know the parents' genetics. Usually, we have genetic data from children and their parents, but we are missing information about grandparents and earlier ancestors. This project explores whether different machine learning models can work backward and fill in those missing pieces.
The system uses msprime to simulate realistic families with known genetic data for everyone. Then, we hide the genetics of some ancestors (simulating missing data) and train five different models to predict those hidden genetics from the visible data. The models are: a Bayesian Model, a Hidden Markov Model, a Deep Neural Network, a Graph Neural Network, and Logistic Regression as a baseline.
Each model predicts the allele dosage (how many copies of a genetic variant someone has: 0, 1, or 2) for each hidden ancestor at each location in the genome. We then compare these predictions to the true values and measure how well each model performs.
Dataset
Choose a dataset
Model
Choose a model
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Conclusion

There is a lot more research that needs to be done in this idea but my results show some promise. The best model is 2x more accurate than random guessing (⅓).

Because stochastic data isn't just genetics, this can be applied beyond genetic ancestral reconstruction but to larger problems.

Questions?

Check out the live site at: capstone.makennaworley.com